RIA PATEL

Biomedical Engineer | BEng (Hons) Biomedical Engineering, First Class
• 07397 183 360 • riapatel1000@icloud.com • linkedin.com/in/riap07

PROFILE

First-class Biomedical Engineering graduate who takes an idea from a CAD model through simulation to a tested prototype. I combine hands-on design, analysis, and testing with strong communication skills from years of customer-facing work. Practical and curious, I enjoy figuring out why things do or don't work.

CAD and design • FEA and simulation • Prototyping and testing • MATLAB • Technical drawings • Customer communication

KEY PROJECTS

Prosthetic Socket Design: Final-Year Dissertation | *Fusion 360, SimScale FEA, FDM 3D printing* 2025 – 2026

- Investigated whether FDM-printed thermoplastics (PA12, PETG, and polypropylene) are suitable for transtibial prosthetic socket components.
- Developed the CAD geometry in Fusion 360, ran structural FEA in SimScale, then 3D-printed prototypes and mechanically tested them.
- Test results agreed closely with the simulation, and PA12 proved to be the best-fitting material.

Plant Keeper: Automatic Plant-Watering System | *Microcontroller, C/C++, Blynk, Fusion 360* 2024

- Programmed a C/C++ microcontroller to read a soil moisture sensor, switch a water pump when needed, and send live data to a Blynk dashboard, then tested and debugged the system.
- Designed a waterproof enclosure in Fusion 360 using Design for Manufacturing (tolerances, fit, assembly sequence, access).
- Produced 2D drawings and a bill of materials, and 3D printed a prototype to check the fit.

Histology Image Analysis Pipeline | *MATLAB* 2026

- Built a MATLAB pipeline that turns stained tissue images into measurable data for my Medical Image Processing module.
- Cleaned the images, separated the haematoxylin and eosin stains using colour deconvolution, and segmented nuclei and stroma with thresholding.
- Measured the results, converting raw microscope images into quantitative data.

EDUCATION AND CERTIFICATIONS

BEng (Hons) Biomedical Engineering, First Class | *Birmingham City University* 2023 – 2026

- Certified SolidWorks Associate (CSWA) 2026 – CSWA Certificate
- Student Member, Institution of Engineering and Technology (IET), since 2024.

A-Levels (English Martyrs Sixth Form): Maths, Biology, Chemistry. GCSEs (Babington Academy): Triple Science, English, Maths, grades above 7.

EXPERIENCE

Customer Retention Advisor | *Hastings Direct* Aug 2026 – Present

- Take around 30 calls a day on renewals, policy changes and cancellations, listening carefully, explaining options clearly and saving customers where I can; meeting or exceeding my retention target with consistently good quality feedback.
- Use Guidewire to amend policies and log detailed notes; completed company training and FCA assessments.
- Support complaints and vulnerable customers by slowing down, following set procedures and arranging warm transfers.

Customer Assistant (Fragrance) | *Superdrug (transferred to Savers)* May 2024 – July 2026

- Advise customers on fragrance and recommend products that suit them, and handle deliveries, stock, and merchandising.
- Stayed reliable through the store's change from Superdrug to Savers in January 2025.

Sales Advisor | *Rituals (John Lewis)* Oct – Dec 2023

- Gave tailored advice in a busy beauty department and ranked in the top three sales performers over the Christmas period.

Production Assistant | *Run & Fly* Jun – Sept 2021

- Analysed the garment cutting workflow to identify bottlenecks and introduced a more streamlined sequence that improved production efficiency.
- Supported quality control inspections in a fast-paced manufacturing environment, contributing to a 10% rise in recorded customer satisfaction.

Earlier roles: Edwards & Miles Eyecare (receptionist placement), Matrix Alliance (brand ambassador), Flair Agency (bar staff).

SKILLS

- **Technical:** Fusion 360, SolidWorks (CSWA), SimScale FEA, MATLAB, C/C++, Excel, basic SQL, FDM 3D printing, hand and power tools.